Full-state feedback control with pole replacement and state-observer

Carried out by

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1. **Purpose of Task:** Create a state space model for the front and rear suspension systems of a two-seat sports car. Design the state-feedback control state observer in MATLAB/Simulink.
2. **Detailed explanation of the realization**

The suspension system is simplified in Figure 1. Where we have the mass of the chassis m_c, the mass of the wheel m_k, the suspension system system explained with a spring and damper; the spring stiffness k_r and the damping constant b_r, the actuator force is neglected for this modeling, the resistance between the ground and wheel is shown as a spring with stiffness k_k with an initial force $u(t)$.

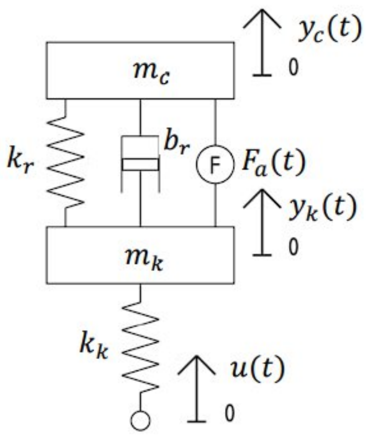

Figure 1: Simplified suspension system model

Using Newton-Euler and Lagrangian method, equations of a simplified suspension system were realized.

<u>Newton-Euler Method</u>

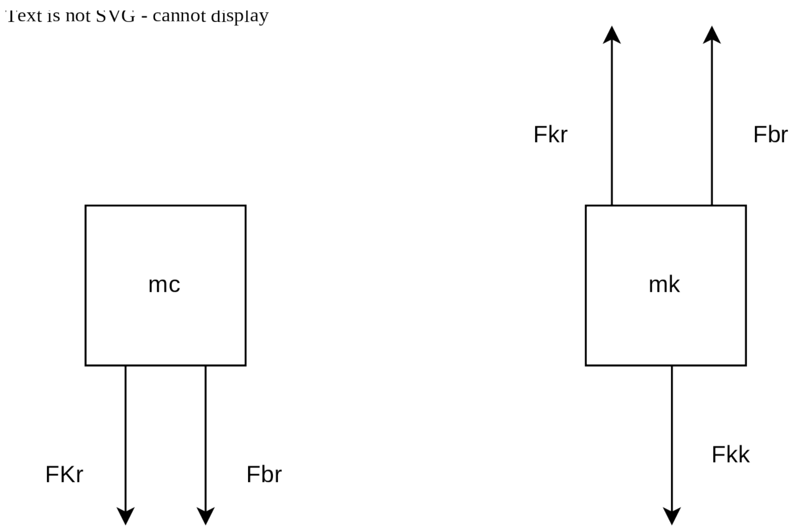

Figure 2: Full body diagram (part a and b from left to right)

For the part a, the equation gotten is in equation 1 below and the part b derived as equation 2.

$$m_c\ddot{y}_c = b_r(\dot{y}_k - \dot{y}_c) + k_r(y_k - y_c) \tag{1.}$$

$$m_k\ddot{y}_k = k_k(u - y_k) - b_r(\dot{y}_k - \dot{y}_c) - k_r(y_k - y_c) \tag{2.}$$

Lagrangian Method

Using the Lagrangian general equation, the equations were realized.

$$\frac{d}{dt}\left(\frac{\boldsymbol{\delta} T}{\boldsymbol{\delta}\dot{q}_i}\right) - \frac{\boldsymbol{\delta} T}{\boldsymbol{\delta} q_i} + \frac{\boldsymbol{\delta} D}{\boldsymbol{\delta}\dot{q}_i} + \frac{\boldsymbol{\delta} U}{\boldsymbol{\delta} q_i} = Q_i$$

The general variable $q_i = y,\ \dot{q}_i = \dot{y},\ \ddot{q}_i = \ddot{y}$

$$T = {}^1\!/_2\, m_c\dot{y}_c^2 \tag{3.}$$

$$D = {}^1\!/_2\, b_r(\dot{y}_k - \dot{y}_c)^2 \tag{4.}$$

$$U = {}^1\!/_2\, k_r(y_k - y_c)^2 + {}^1\!/_2\, k_k(U - y_k)^2 y_k \tag{5.}$$

When $i = c$,

$$\frac{\delta T}{\delta\dot{q}_c} = m_c\dot{y}_c,\ \frac{d}{dt}\left(\frac{\delta T}{\delta\dot{q}_c}\right) = m_c\ddot{y}_c,\ \frac{\delta D}{\delta\dot{q}_c} = -b_r(\dot{y}_k - \dot{y}_c),\ \frac{\delta U}{\delta q_c} = -k_k(y_k - y_c)$$

$$m_c\ddot{y}_c - b_r(\dot{y}_k - \dot{y}_c) - k_r(y_k - y_c) = 0 \tag{6.}$$

When $i = k$,

$$\frac{\delta T}{\delta\dot{q}_k} = m_k\dot{y}_k,\ \frac{d}{dt}\left(\frac{\delta T}{\delta\dot{q}_k}\right) = m_k\ddot{y}_k,\ \frac{\delta D}{\delta\dot{q}_k} = b_r(\dot{y}_k - \dot{y}_c)\,y_k$$

$$\frac{\delta U}{\delta q_k} = k_r(y_k - y_c) - k_k(u - y_k)$$

$$m_k\ddot{y}_k + b_r(\dot{y}_k - \dot{y}_c) - k_r(y_k - y_c) + k_k y_k = k_k u \tag{7.}$$

Graphical modelling

The graphical modeling includes a bond graph, a block diagram and a system loop diagrams. The flow of model is identified as the velocity $\dot{y}$ and for effort as the force. The masses are the I elements, the springs are the C elements and the damper is the R element. The initial force is decided to be a source of effort.

The bond graph, block diagram and system diagrams are in the figures below.

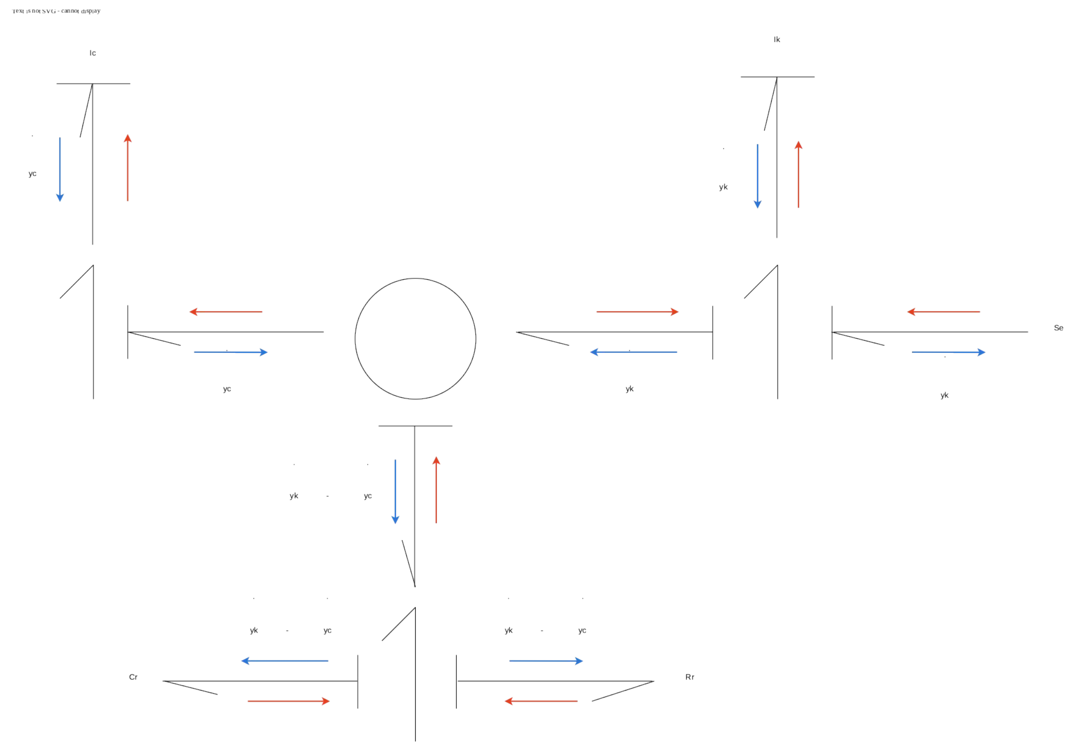

Figure 3: Bond graph of model

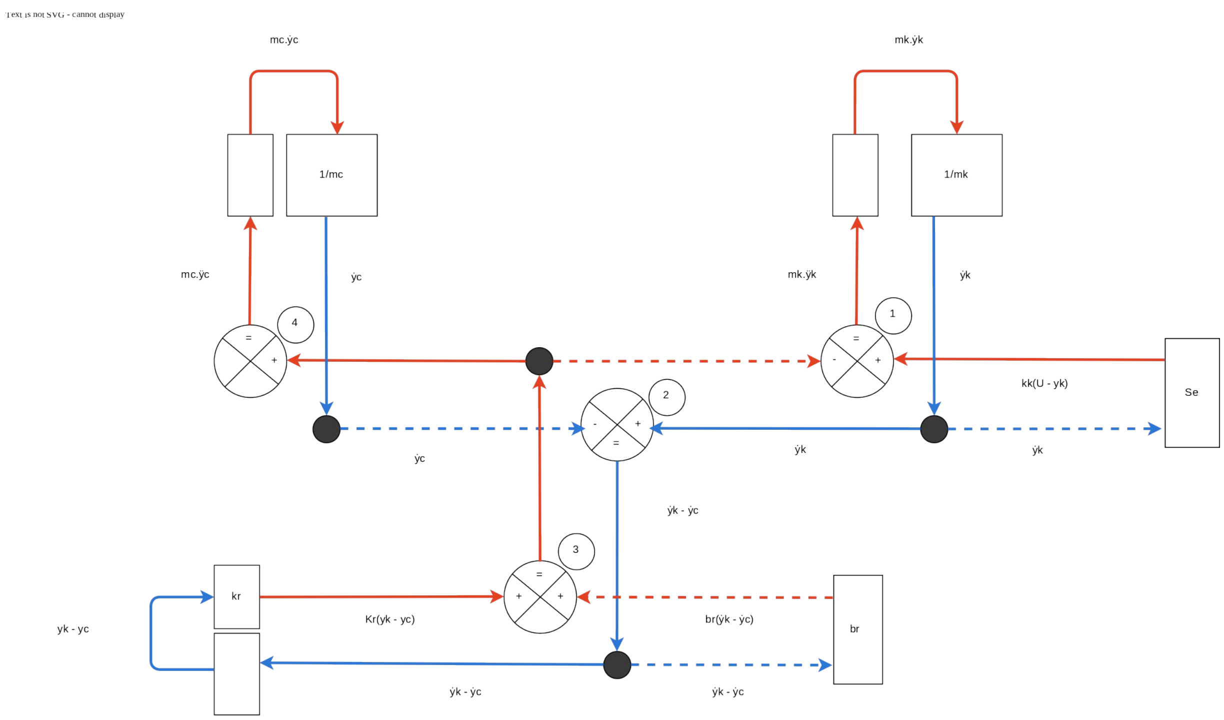

Figure 4: Block diagram

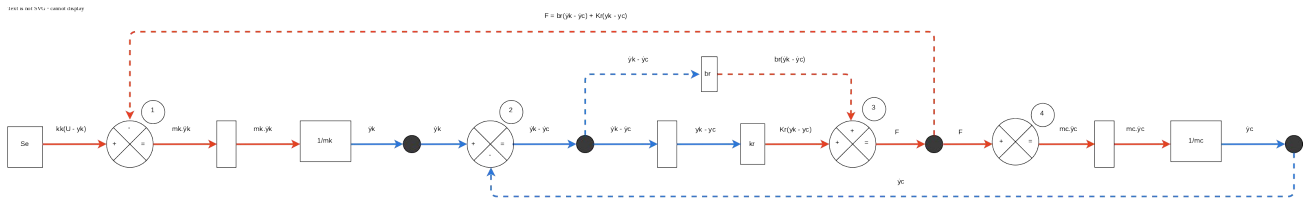

Figure 5: System loop

From the the system loop diagram, summation icon 1 gives the same equation 2 and summation icon 4 gives the same equation 1, thus supporting the Lagrangian and Newton-Euler models.

By rearranging equations, we get the forms below.

$$\ddot{y}_k = \left(\frac{k_k}{m_k}\right)u - \left(\frac{k_k+k_r}{m_k}\right)y_k + \left(\frac{k_r}{m_k}\right)y_c - \left(\frac{b_r}{m_k}\right)\dot{y}_k + \left(\frac{b_r}{m_k}\right)\dot{y}_c \qquad (8.)$$

And from equation 2, we get the form below.

$$\ddot{y}_c = \left(\frac{k_r}{m_c}\right)y_k - \left(\frac{k_r}{m_c}\right)y_c + \left(\frac{b_r}{m_c}\right)\dot{y}_k - \left(\frac{b_r}{m_c}\right)\dot{y}_c \qquad (9.)$$

State Space Model

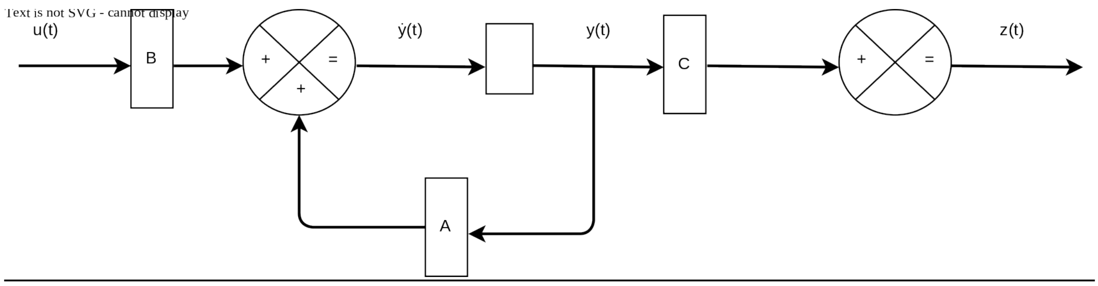

Figure 6: State space model

Using the general state equation and the system model equations 8 and 9, the state equation is written as.

$$\dot{y}(t) = A.y(t) \; + B.u(t)$$

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}\frac{d}{dt} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{-k_k-k_r}{m_k} & \frac{k_r}{m_k} & -\frac{b_r}{m_k} & \frac{b_r}{m_k} \\ \frac{k_r}{m_c} & -\frac{k_r}{m_c} & \frac{b_r}{m_c} & -\frac{b_r}{m_c} \end{bmatrix} \cdot \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{k_k}{m_k} \\ 0 \end{bmatrix} .u(t) \qquad (10.)$$

Where $y_1 = y_k,\; y_2 = y_c,\; y_3 = \frac{dy_1}{dt} = \dot{y}_1,\; y_4 = \frac{dy_2}{dt} = \dot{y}_2$

And an output equation

$z(t) = C.y(t) \; + \; D.u(t)$

The D matrix has a value of 0 and for this model, we would be looking at only one output. That is, the displacement of the chassis from the ground y_c .

$$z_1 = [0 \quad 1 \quad 0 \quad 0] . \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} \qquad (11.)$$

Modeling and Control in MATLAB

The full-state control and state space observer of the system are modelled in MATLAB based on Figure 7 and Figure 8. By substituting the values for the variables, the poles for the system characteristic equation was gotten and edited to preferable values to avoid overshoot and achieve a steady while keeping the characteristics of the system.

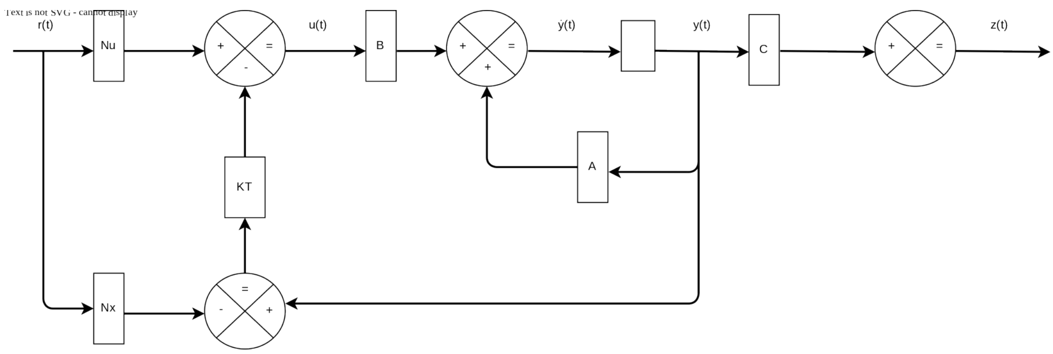

Figure 7: Full-state Feedback control

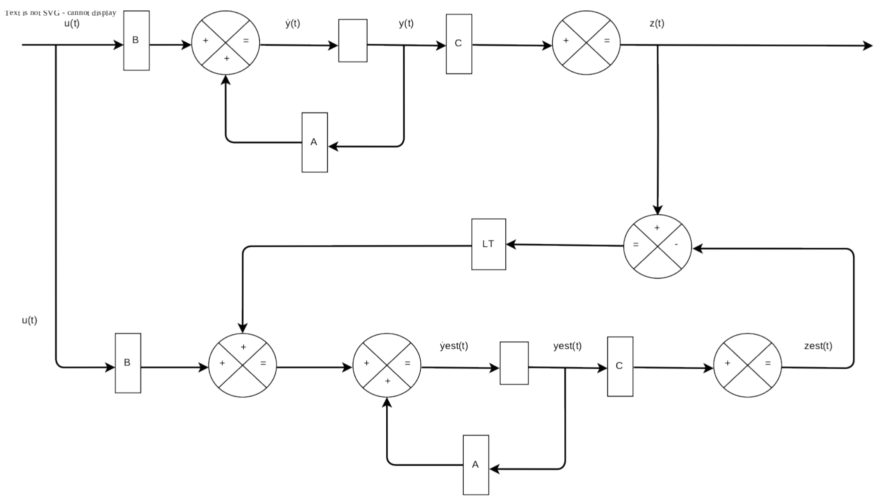

Figure 8: State space Observer

For the FRONT variables:

- Using the controllability canonic method, the amplification vector K^T is derived.

– Both characteristic equations are compared to derive the amplification vector.
– This is repeated using the Ackermann formula.
– The full state feedback control compensation variables are derived.
– The controllability compensation vector for the observation is derived.

This process is repeated for the REAR variables values substituted in the state space model matrices.

3. Results

Full-state feedback control

In Simulink, the full-state feedback control is modelled. For the front, the control model is as shown in Figure 9 and the result is shown in Figure 10.

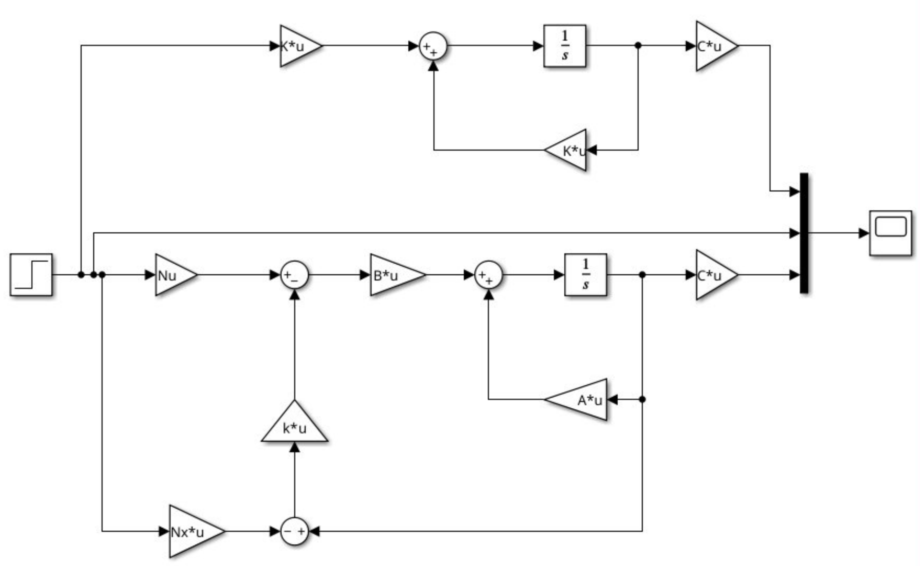

Figure 9: Full state feedback control model

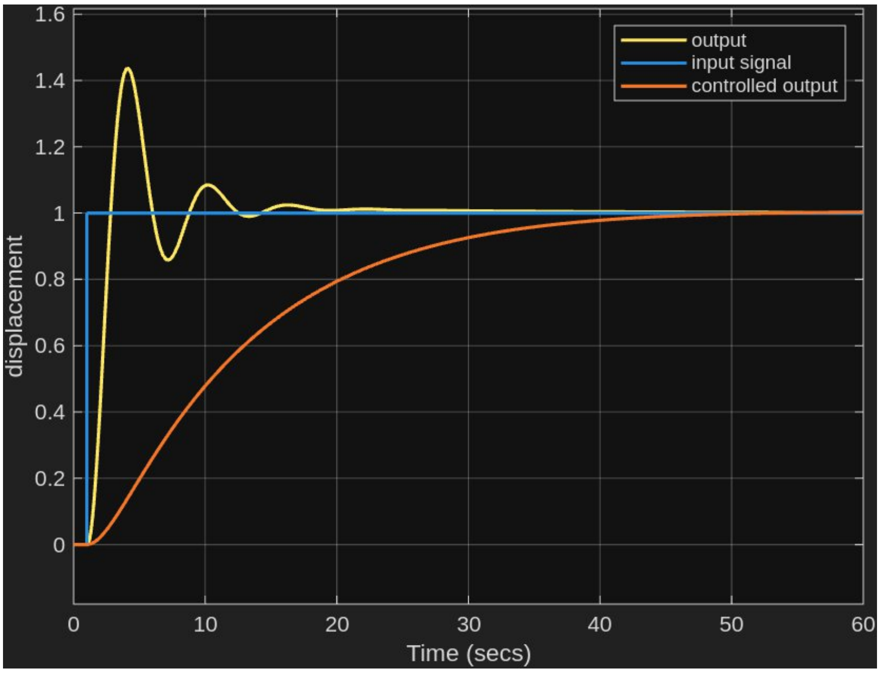

Figure 10: Front Full-state control model graph

The system is controlled with no overshoot and settles around the same time as the uncontrolled system at about 55 seconds.

And for the rear, the control model is done with same model in Figure 9 and the result is shown in Figure 11.

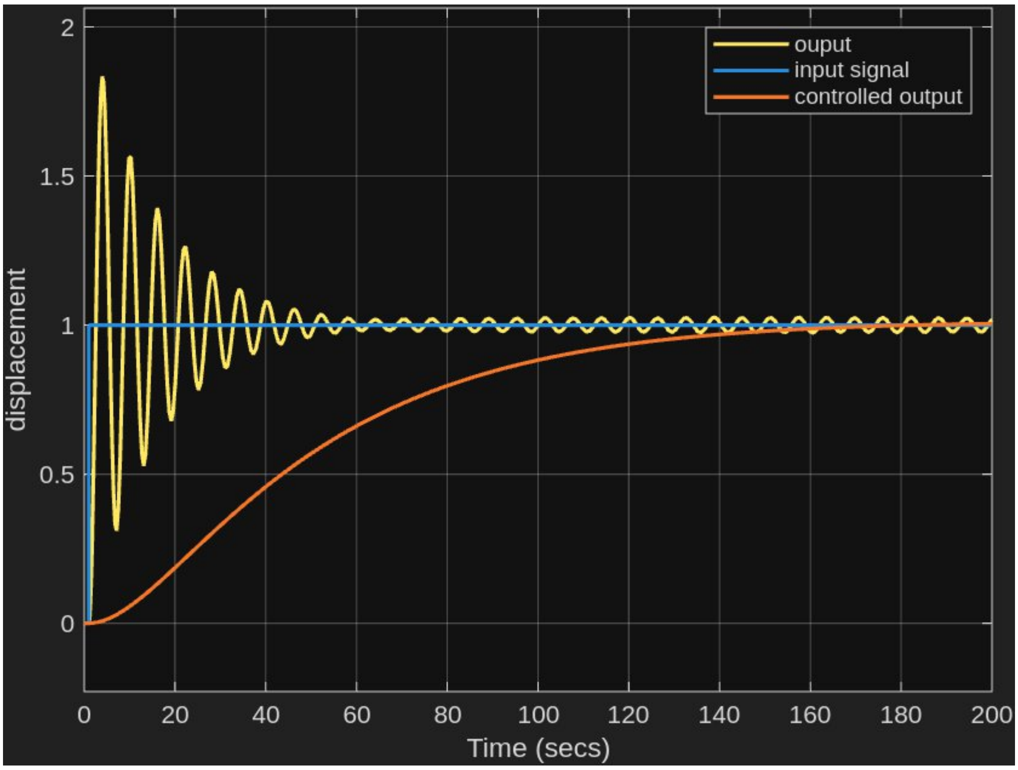

Figure 11: Rear Full-state control model graph

The system is controlled and achieves a steady state at around 180 seconds.

State Space Observer

The state space observer is modelled in Simulink as well. The front model is shown in Figure 12 and the result in Figure 13.

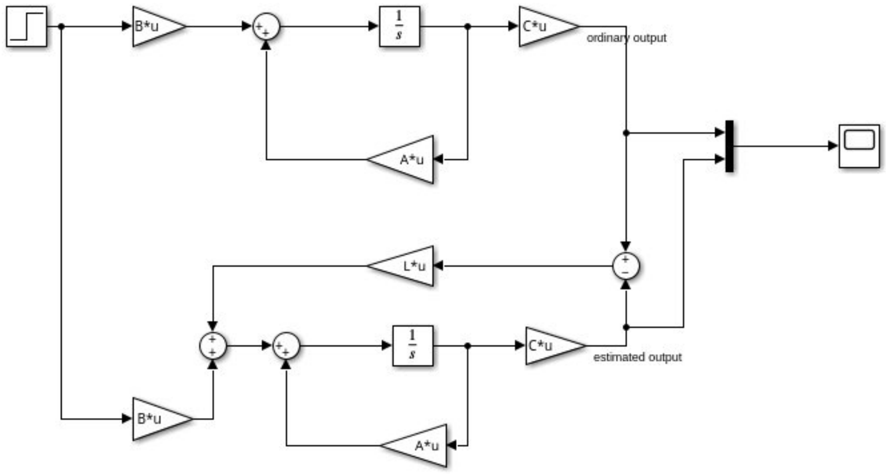

Figure 12: Full-state Observer model

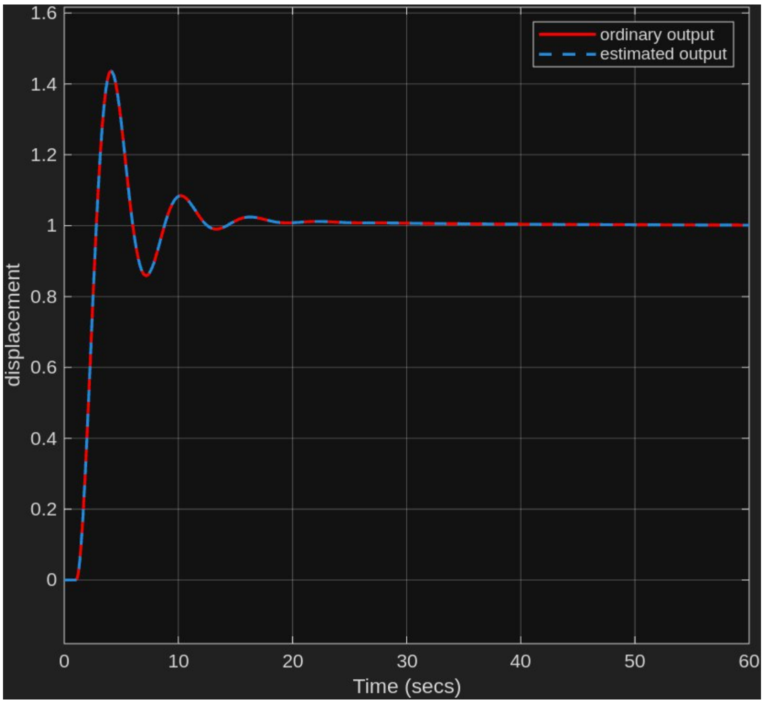

Figure 13: Front observer model graph

The estimated and normal system overlap.

And for the rear model shown in Figure 14, with an additional scope and the results in Figure 15 and Figure 16.

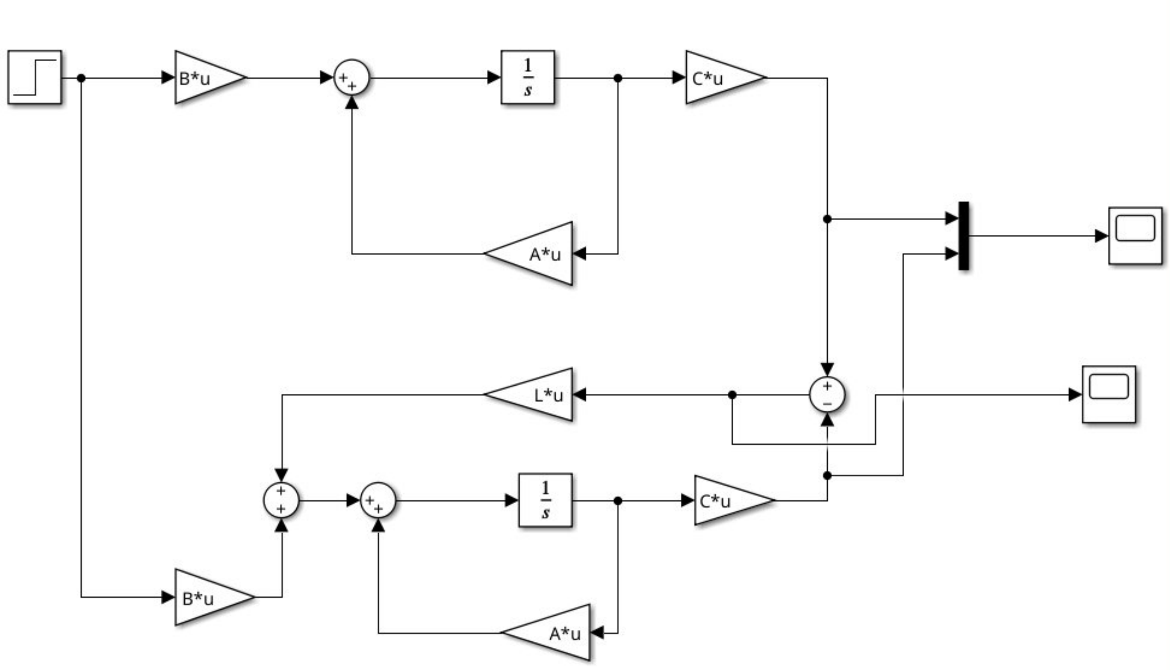

Figure 14: Rear Full state observer model

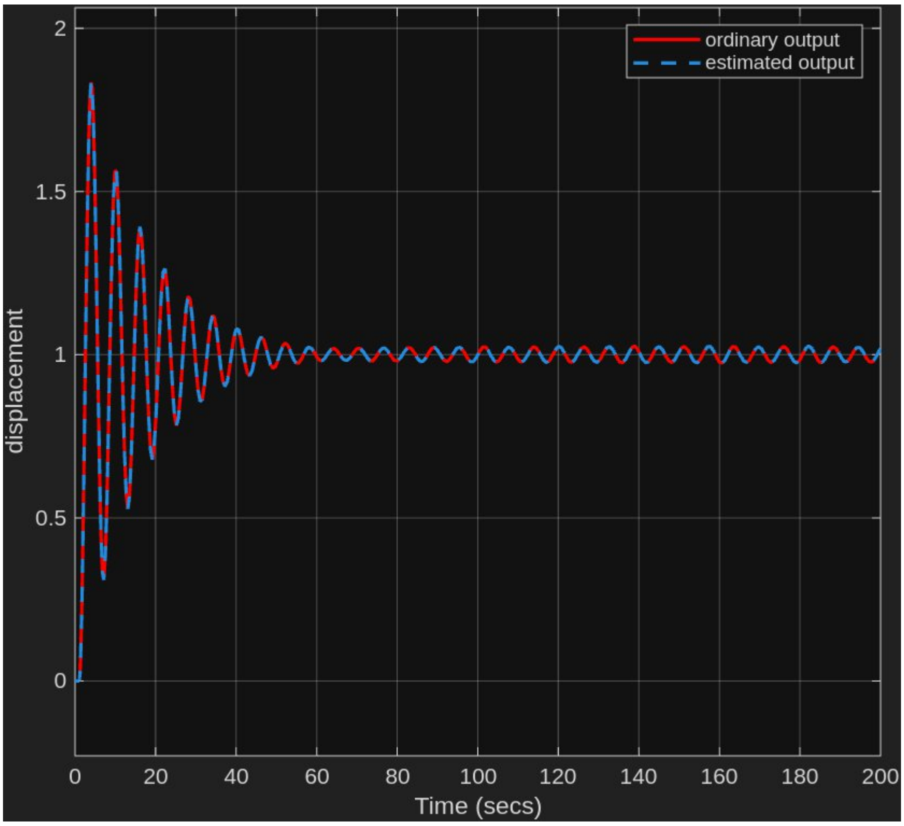

Figure 15: Rear Observer Model graph

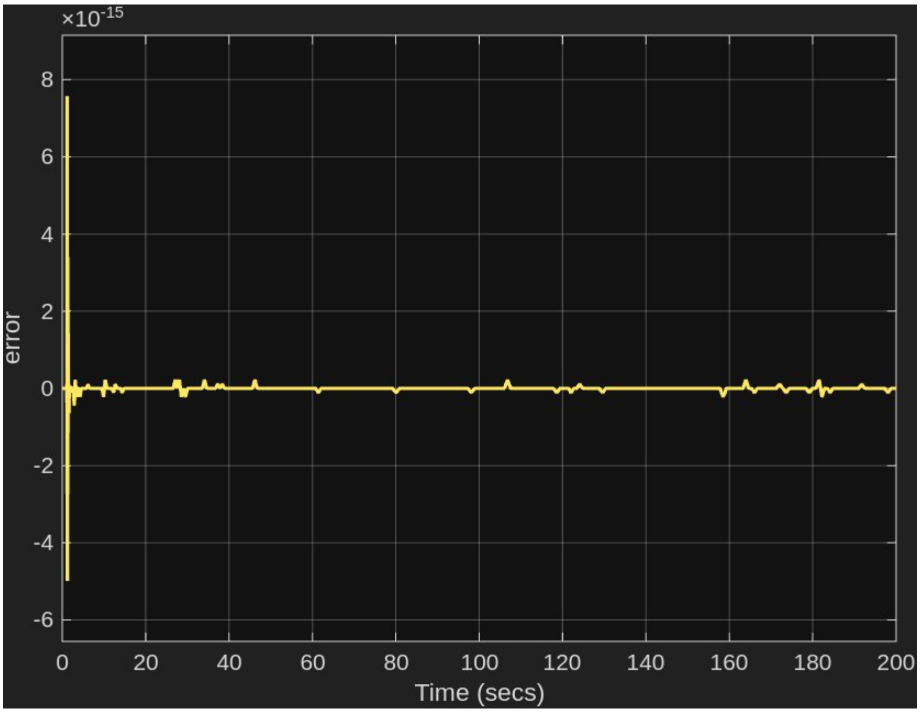

Figure 16: Rear Observer Error graph

The ordinary and estimated signals overlap. From the error graph, it helps understand that regardless of the oscillation still occurring as seen in the observer graph, the error is close to or at 0.

Using Simulink , a pothole and speed-bump scenarios were visualized for the front and rear suspension systems.

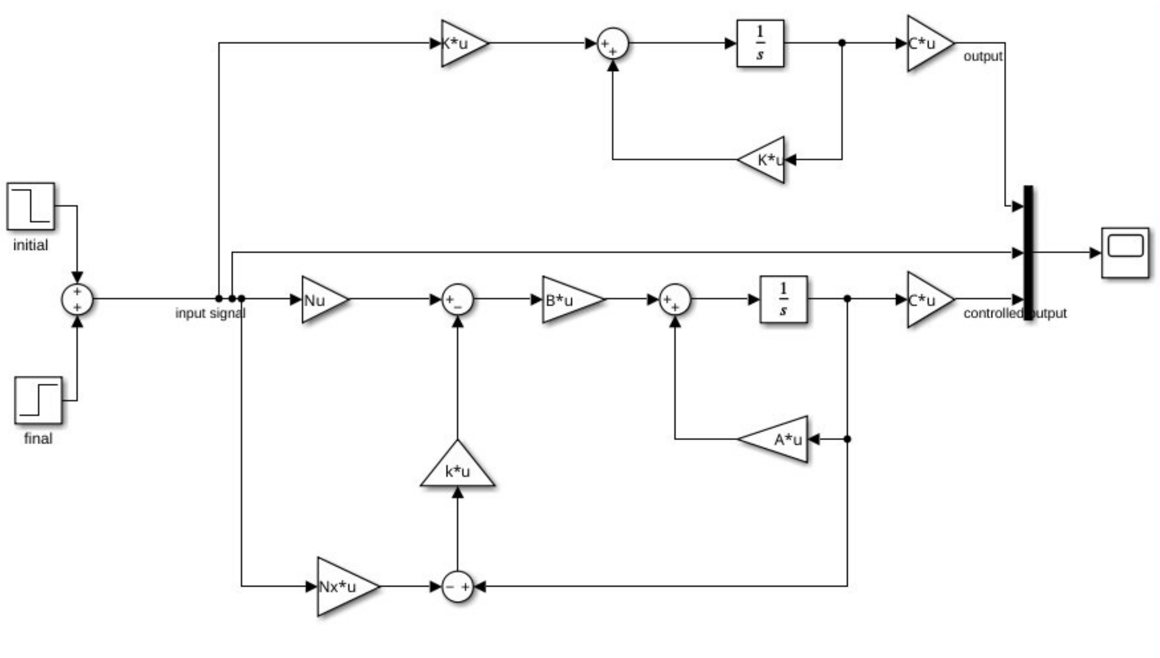

Figure 17: Pothole and Speed-bump scenario Full-state control

<u>Scenario 2</u>

There is a pothole on the road that the wheel covers in 10 seconds (t1 = 5s and t2 = 15s), with the depth of 2 units.

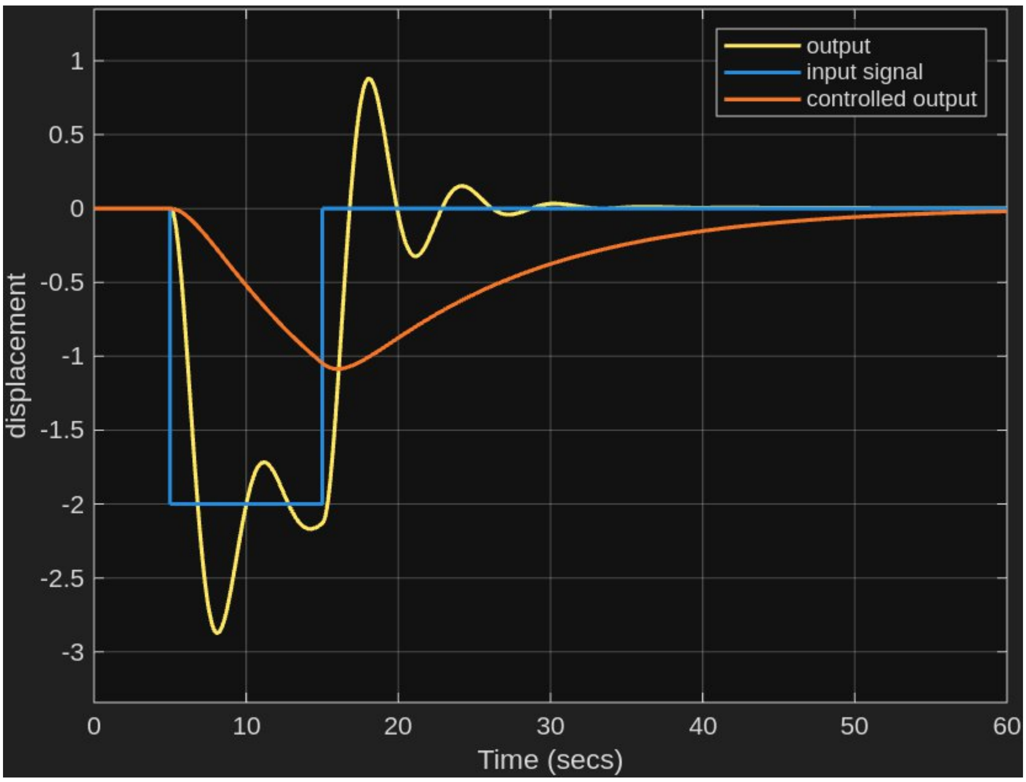

Figure 18: Front pothole reaction graph

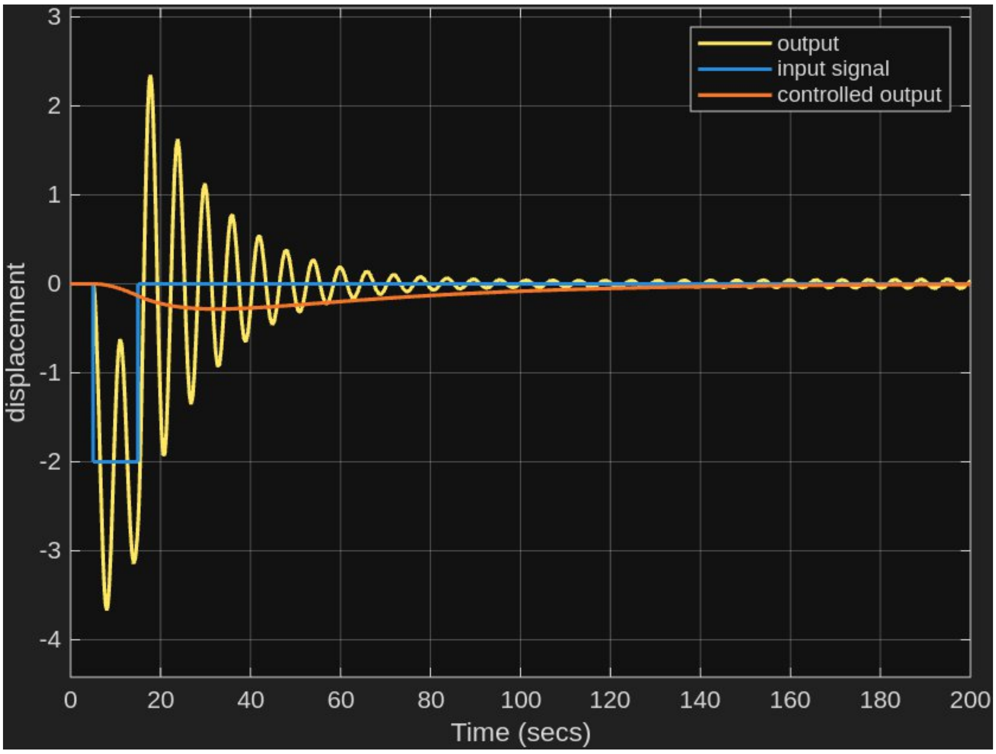

Figure 19: Rear pothole reaction graph

The controlled front suspension system provides a smooth entry and exit to and from the pothole and goes to a steady state within 57 seconds of the pothole encounter. For the rear, displacement does not fall as much as the front and it gets to a steady state in 155s from the exit of the pothole.

<u>Scenario 3</u>

There is a speed-bump on the road that the wheel covers in 1 seconds (t1 = 5s and t2 = 6s), with the height of 8 units.

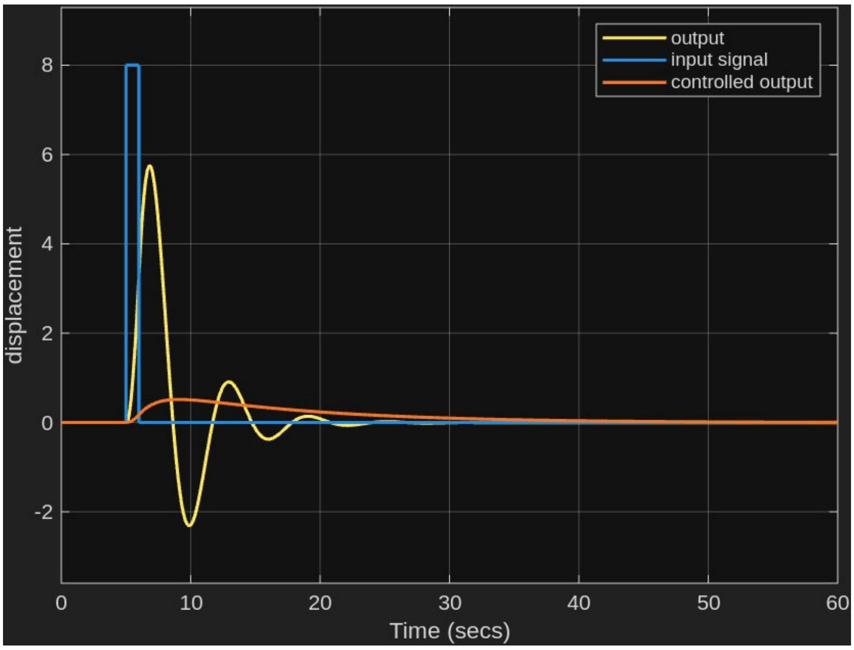

Figure 20: Front speed-bump reaction

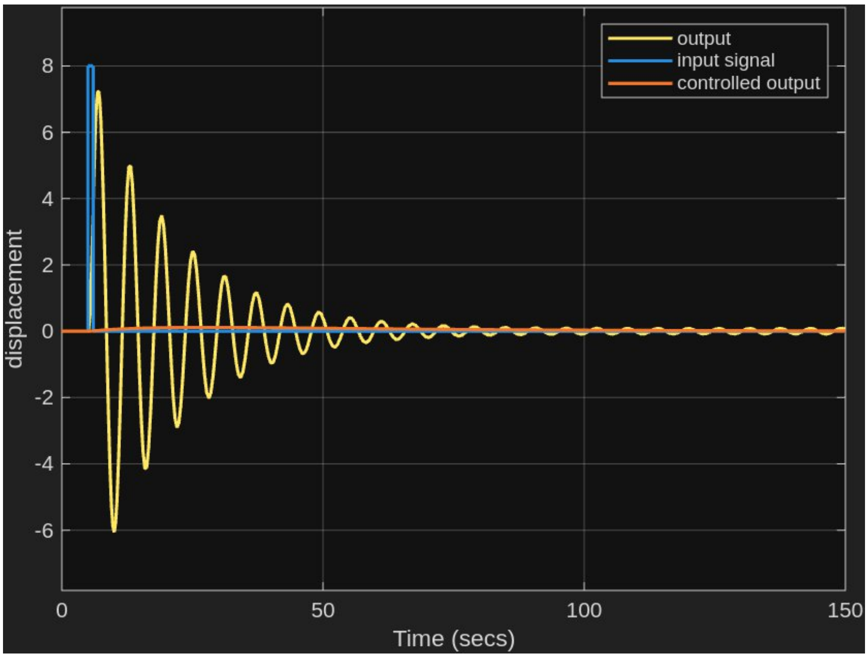

Figure 21: Rear speed-bump reaction

The controlled front suspension system shows a sharp increase in displacement on contact with the speed-bump and gets to steady state around time of 55 seconds. The rear shows a small increase in displacement and gets to a steady state before time 150s.

4. Conclusion

The front suspension system has a transient time of around 55 seconds and the rear suspension system has a transient time of around 180 seconds in a scenario of steady

excitement.
In the original scenario we can assume that when a change in ground level at 1 unit with no reversal, the front and rear suspension systems would take around 55 and 180 seconds respectively for the displacement of the chassis to return to normal and settle.

In the cases of the pothole and the speed-bump, both systems arrive steady state faster in the speed-bump scenario. The rear system shows little change in displacement compared to the front on both scenarios, this could be due to the high damping coefficient value.